1. Introduction to Immersive Technologies (0–10 minutes)

What is Reality?

- Human perception of the **physical world** through senses.
- Limitations of natural perception.

Why AR/VR/MR?

- Extend, enhance, or replace real-world perception.
- Enable **simulation, training, visualization, and interaction**.

Definitions:

- **Virtual Reality (VR):**
  A fully computer-generated environment that replaces the real world and allows user interaction.
- **Augmented Reality (AR):**
  Overlays digital information (text, images, 3D objects) onto the real-world environment.
- **Mixed Reality (MR):**
  Combines real and virtual worlds where digital objects interact with the real environment in real time.

📌 Example:

- VR: Flight simulator
- AR: Pokémon Go
- MR: Microsoft HoloLens applications

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2. Evolution of AR, VR, and MR (10–20 minutes)

Early Concepts:

- 1838 – Stereoscope (Charles Wheatstone)
- 1950s – Sensorama (Morton Heilig)

Development Phase:

- 1968 – First Head-Mounted Display (Ivan Sutherland)
- 1990s – VR used in military and research labs

Modern Era:

- 2012 – Oculus Rift revival
- 2016 – Pokémon Go popularizes AR
- 2020+ – Industrial, medical, and educational MR systems

📌 **Discussion Point:**
Why did AR/VR take decades to become mainstream?

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3. Reality–Virtuality Continuum (20–35 minutes)

Concept:

Proposed by **Milgram & Kishino (1994)**

Continuum Levels:

1. **Real Environment**
2. **Augmented Reality (AR)**
3. **Mixed Reality (MR)**
4. **Augmented Virtuality**
5. **Virtual Reality (VR)**

Explanation:

- As we move from left to right, **virtual content increases**.
- MR lies between AR and VR.

📌 **Diagram Explanation (show slide):**

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Real World —— AR —— MR —— Augmented Virtuality —— VR
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📌 **Use Case Mapping:**

- Google Maps AR → AR
- HoloLens → MR
- Meta Quest → VR

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4. Milestones in AR/VR History (35–50 minutes)

Key Milestones:

- **1968:** Sword of Damocles (first HMD)

- **1994:** Reality–Virtuality Continuum defined
- **2012:** Oculus Rift Kickstarter
- **2016:** Pokémon Go
- **2019:** Microsoft HoloLens 2
- **2023+:** Metaverse & AI-powered XR

Impact:

- Shift from research labs → consumer & industry adoption

📌 **Case Snapshot:**
Healthcare using VR for surgical training

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5. Comparison Overview: AR vs VR vs MR (50–55 minutes)

Feature	AR	VR	MR
Real World Visible	Yes	No	Yes
Immersion Level	Low	High	Medium–High
Devices	Mobile, Glasses	Headsets	HoloLens
Interaction	Limited	Full	Advanced

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6. Recap & Student Interaction (55–60 minutes)

Quick Recap:

- AR adds to reality
- VR replaces reality
- MR blends both
- Continuum explains transition

🎬 **Explain Augmented Reality (AR), Virtual Reality (VR), and Mixed Reality (MR). Compare them with suitable examples.**
(Long answer – 8 to 10 marks)

🎬 **Describe the Reality–Virtuality Continuum proposed by Milgram and Kishino. Explain each stage with applications.**
(Long answer – 8 marks)

🎬 **Trace the evolution and major milestones in the development of AR and VR technologies.**
(Short/Medium answer – 5 marks)